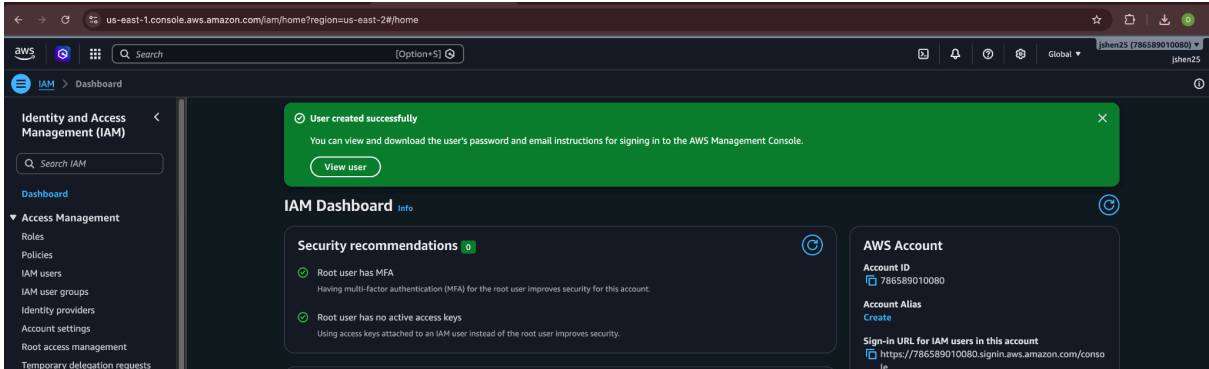


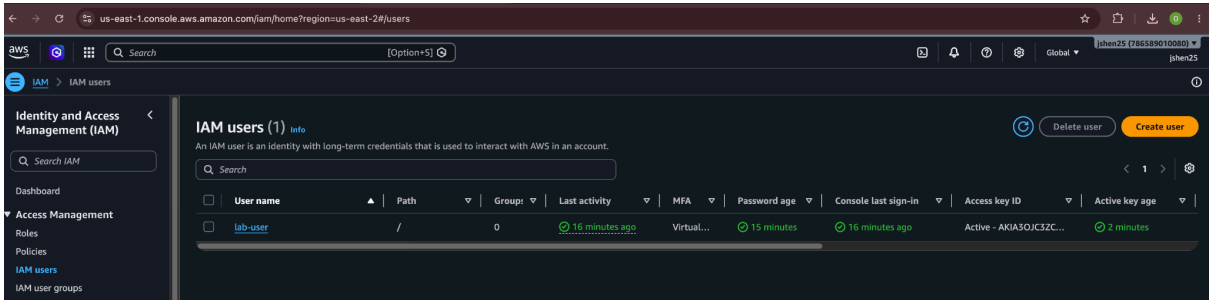
Lab 01

Checklist?

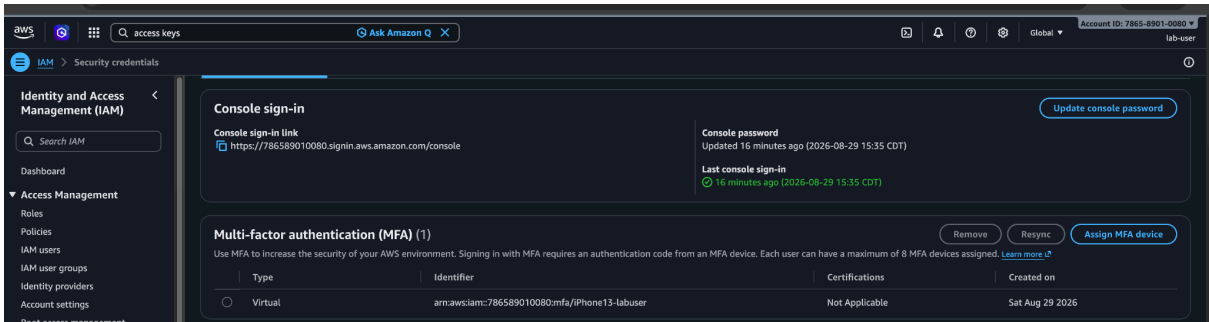
IAM MFA



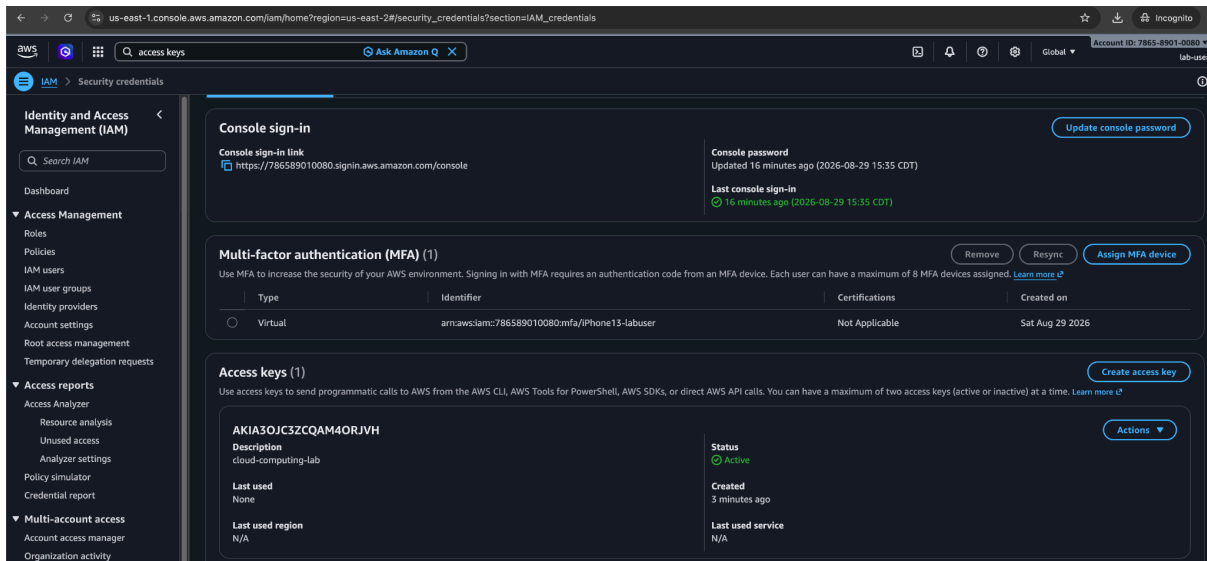
Users:



Users MFA:



Users access key:



Root user MFA and least-privilege IAM users are important for cloud security. The root user has full control of the cloud account, so if someone steals its password, they can cause very serious damage. MFA gives one more security step to protect this account. IAM users should only get the permissions they need for their jobs. This can reduce security risks, prevent mistakes, and limit the damage if one user account is hacked.

Lab02 :

docker --version

```
jieshen — jieshen@MacBookPro — — -zsh — 167x43
Last login: Fri Aug 28 01:28:36 on console
[~] ~ docker --version
Docker version 29.7.2, build a7dcaa6
```

hello-world

```

Last login: Fri Aug 28 01:28:36 on console
➤ ~ docker --version
Docker version 29.7.2, build a7dcaa6
➤ ~ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

```

inside:

```

root@12a14a8e9a4d:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
root@12a14a8e9a4d:/# whoami
root
root@12a14a8e9a4d:/# apt update

```

ps -a

```

➤ ~ docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
12a14a8e9a4d	ubuntu:22.04	"/bin/bash"	2 minutes ago	Exited (0) About a minute ago		my-ubuntu
8c367dd443c7	hello-world	"/hello"	3 minutes ago	Exited (0) 3 minutes ago		nifty_benz
98b5a8b996cb	sheep-sheep-frontend	"python3 app.py"	9 months ago	Exited (0) 9 months ago		sheep-sheep-frontend-run-2526b261e8ff
d0702f8f60b7	sheep-sheep-server	"applications/run-sh_"	9 months ago	Exited (137) 9 months ago		sheep-sheep-server_1
1ed0e4d4f5d0	pintos_image	"/bin/bash"	15 months ago	Exited (0) 14 months ago		pintos
7dc176f9f9bc	b532ce31b4f9	"/bin/bash"	15 months ago	Exited (0) 15 months ago		interesting_neumann
c2f2522b8cd0	hello-world	"/hello"	15 months ago	Exited (0) 15 months ago		thirsty_boyd
5a1cabfb6f3e	hello-world	"/hello"	15 months ago	Exited (0) 15 months ago		heuristic_noyce

```

Total reclaimed space: 113.8MB
➤ ~ docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
➤ ~						
➤ ~						
➤ ~						
➤ ~						
➤ ~						
➤ ~						

Docker Image:

Docker Container:

in my understanding, Image is just like Windows/Ubuntu ISO software binary, container is a virtual machine generated or installed from the Image. Images can be downloaded from internet. Container is build in our local machine, Mac or Win or Linux.

Lab3:

aws – version

```
root@e4d554fa1453:/# aws --version
aws-cli/2.36.34 Python/3.14.6 Linux/7.0.12-linuxkit exe/x86_64.ubuntu.22
```

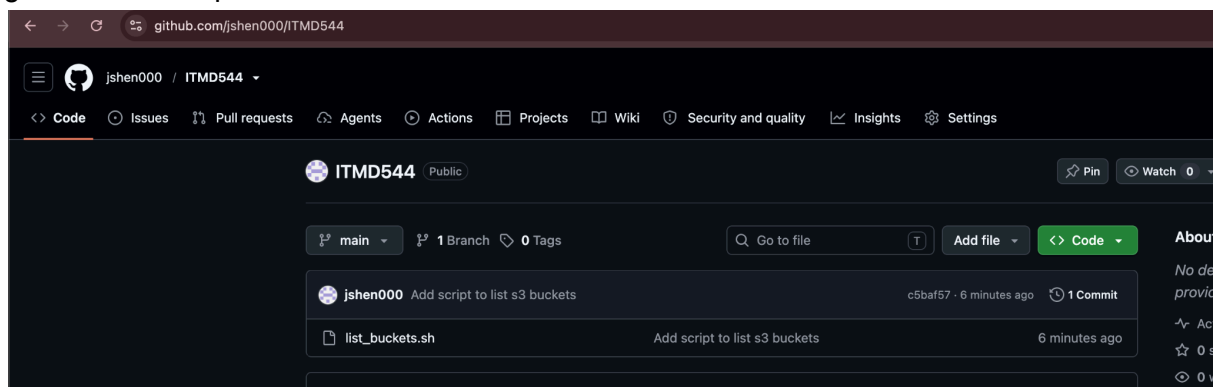
aws sts

```
root@e4d554fa1453:/# aws sts get-caller-identity
{
  "UserId": "AIDA30JC3ZCQMU6RQTW4S",
  "Account": "786589010080",
  "Arn": "arn:aws:iam::786589010080:user/lab-user"
}
```

git config :

```
root@e4d554fa1453:/# git config --global --list
user.name=Jie Shen
user.email=jshen25@hawk.illinoistech.edu
```

github show scripts:



We should never commit AWS credentials or GitHub tokens to a repository, even a private one, because private repositories are not completely safe. Other collaborators, stolen accounts, wrong permissions, or security problems may expose the secrets. Git also keeps commit history, so deleting the credentials later may not fully remove them. If attackers get these credentials, they can access cloud resources, steal data, change code, create expensive services, or delete systems. Credentials should be stored in environment variables or a secure secrets manager instead.

